



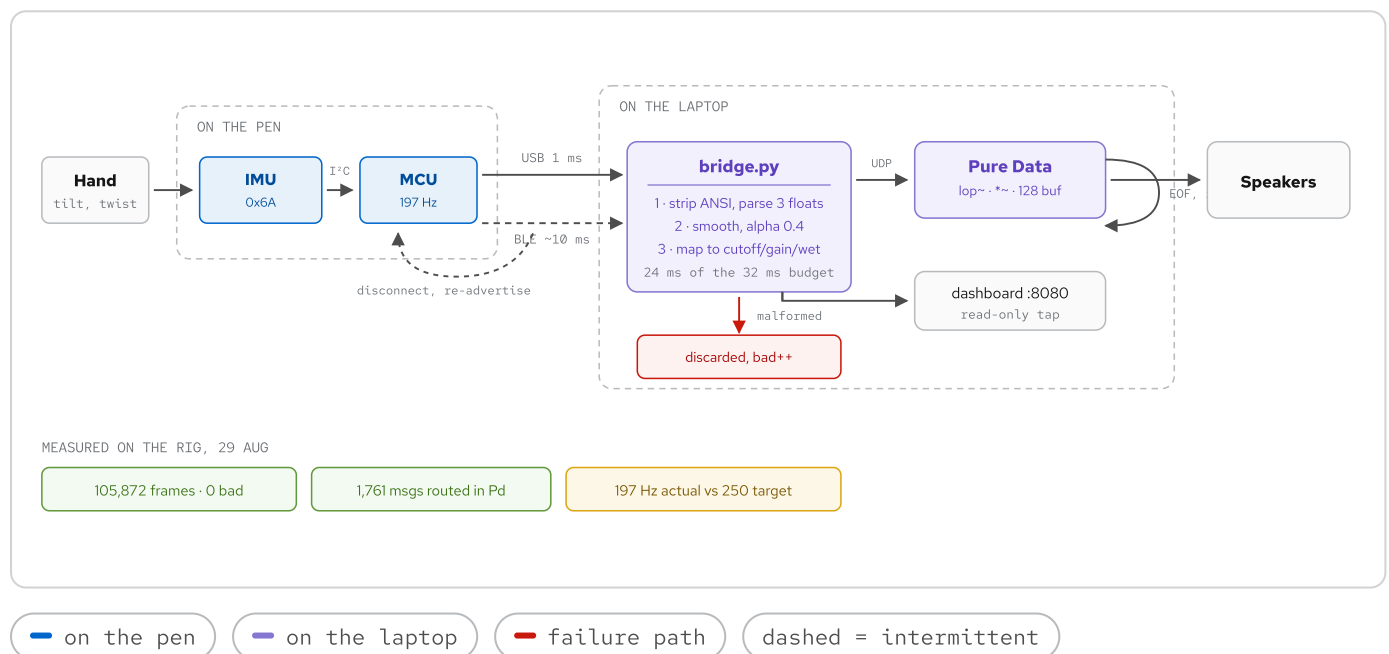
SYSTEM DOCUMENTATION

How a pen ends up controlling a filter

Numbers on this page are measured on the working rig, not estimated. Where something is estimated or unverified it says so.

Signal chain

Two transports, one discard path, and two loops. The loops are the parts that took longest to get right, so they are drawn rather than described.

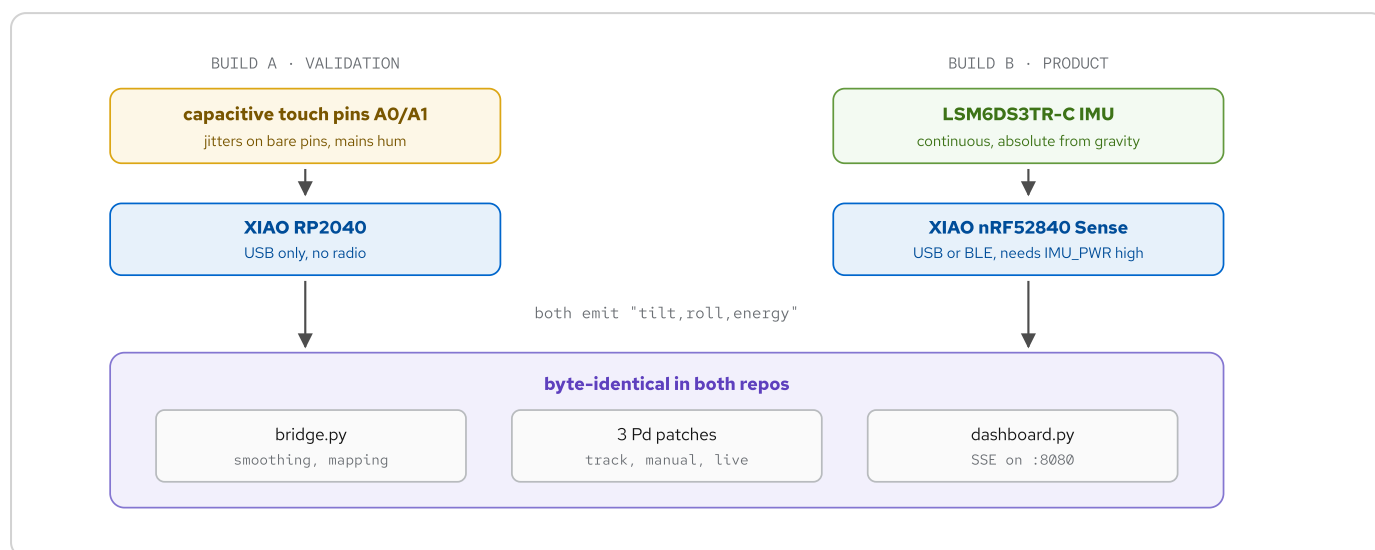


The bridge is drawn largest because it holds all three transforms and most of the tuning. The red edge is the discard path: any frame that does not split into three floats is dropped and counted, and the dashboard shows that counter. Both loops are real. BLE re-advertises after a disconnect, and Pd reopens the file at end of playback, which is the only reason the track repeats.

The 197 Hz gap: USB scheduling plus CircuitPython's `print()` overhead against a 250 Hz target. It has not caused problems, but it means the smoothing figures below settle slightly slower in practice than the table says.

Two builds

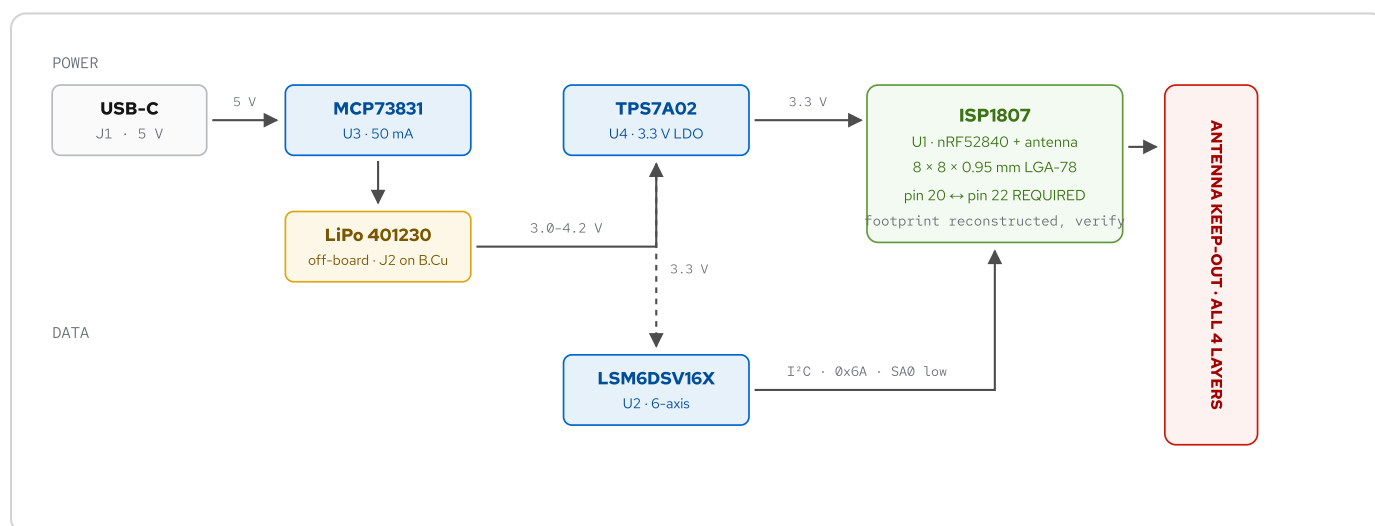
The RP2040 build exists so the chain could be built before the motion sensor arrived. What is worth drawing is the seam: where they differ and where they stop differing.



Moving from validation to product means replacing `read_motion()`. The gold box is gold because touch is a compromise, not a preview: bare pins pick up mains hum, so the values wander even untouched.

Production board

12 x 30 mm, four layers, 0.6 mm, all SMD. Two constraints drive the whole layout, and one of them dictates the board's shape.



The cell cannot feed the module directly: it runs to 4.2 V and the module's ceiling is 3.6 V, so it returns through the LDO. The keep-out is drawn full height because it is the constraint that decides the board's shape, not a footnote to it.

Unverified: the ISP1807 land pattern in `stylus.pretty/` was reconstructed from the datasheet drawing. Pad numbering is read from the datasheet figure and is correct; the coordinates are derived. On a 0.65 mm pitch 78-pad LGA, check it against Insight SiP's official land pattern before paying for fabrication.

Smoothing

Measured settling to 95% of a step at 250 Hz. Past roughly 20 ms end to end the instrument stops feeling connected to your hand.

ALPHA	SETTLES IN	FEEL
0.05	236 ms	unusable
0.15	76 ms	visibly behind your hand
0.4	24 ms	the default
0.5	20 ms	snappier, more jitter through
1.0	0 ms	raw, audible clicks on the filter

Smoothing eats 24 of the 32 ms budget and it is the only term you can change freely. A one-pole filter picks one setting for both fast and slow motion. A one-euro filter adapts instead, and that is the right fix once the mapping stops changing.